Appendices

Moderated Learning Management System: CS4433 Final Project Report

Appendix A: Requirements Traceability Matrix

Req.	Use case	Design elements	Source files	Test cases
FR-01	Log in (included by all)	AuthService, User, IPasswordHasher; table users	AuthService.cpp, User.cpp, SodiumPasswordHasher.cpp	Authentication.LoginSucceeds…, User.AuthenticationChecks…
FR-02	All	User subclasses, Permission, Guards; users.role	User.cpp, Guards.hpp	AccessControl.* (6), User.* (6)
FR-03	Manage users	AdminService; users	AdminService.cpp, SqliteRepositories.cpp	AdminRules.AccountValidation
FR-04	Manage departments, courses	AdminService, ICourseRepository; departments, courses, course_lecturers	AdminService.cpp, SqliteRepositories.cpp	AdminRules.AccountValidation, test harness setup
FR-05	Enrol students	AdminService; enrolments	AdminService.cpp	Test harness setup; AccessControl.UnenrolledStudent…
FR-06	Create assignment	AssignmentService; assignments	AssignmentService.cpp	Workflow.AttemptLimitIsEnforced, harness setup
FR-07	Grant extension	AssignmentService, DeadlinePolicy; extensions	AssignmentService.cpp, DeadlinePolicy.cpp	Workflow.Extension* (2), Penalty.ExtendedDeadline…
FR-08	UC-01 Submit assignment	SubmissionService, Sequence diagram 1; submissions	SubmissionService.cpp	AccessControl.UnenrolledStudent…, Workflow.*
FR-09	UC-01 (alternative flow)	SubmissionService, SubmissionStateMachine	SubmissionService.cpp	Workflow.Resubmission* (2), AttemptLimit, MarkedWork…
FR-10	UC-01, UC-02	ILatePenaltyStrategy, DailyLatePenaltyStrategy, LateResult	DailyLatePenaltyStrategy.cpp, MarkingService.cpp	Penalty.* (13), Workflow.Late*, Workflow.WorkMoreThan…
FR-11	UC-02 Mark submission	MarkingService; marks	MarkingService.cpp	Workflow.InvalidMarksAreRejected, Workflow.MarkChanges…
FR-12	UC-03 Moderate marks	ModerationService; moderation_records	ModerationService.cpp	Workflow.ChangesRequested* (2), AccessControl.HeadOfDepartment…
FR-13	UC-03 Release marks	ModerationService, ResultsService; marks.status	ModerationService.cpp, ResultsService.cpp	Workflow.MarkIsHidden…, AccessControl.ReleaseRequires…, AccessControl.StudentsSee…
FR-14	UC-02, UC-03	AuditService; audit_log (append-only triggers)	AuditService.cpp, schema.sql	Workflow.MarkChanges…, Integrity.AuditLog…
FR-15	All submission use cases	SubmissionStateMachine, state diagram; submissions.status	SubmissionStateMachine.cpp, Submission.hpp	StateMachine.* (6), Types.*
FR-16	View courses and assignments	AssignmentService.listFor, SubmissionService.mySubmissions	AssignmentService.cpp, SubmissionService.cpp	AccessControl.StudentsSeeOnly…
FR-17	Manage users	AdminService.setUserActive; users.active	AdminService.cpp	Authentication.DeactivatedAccounts…
FR-18	Change password	AdminService.changeOwnPassword	AdminService.cpp	Authentication.ChangePassword…
FR-19	View audit history	AuditService.history	AuditService.cpp	AuditTrail.OnlyAdministrators…

Req.	Use case	Design elements	Source files	Test cases
FR-20	Grant extension (class-wide)	AssignmentService.grantClassExtension (CR1)	AssignmentService.cpp (v1.1)	v1.1 tests (see section 10)
NFR-01	Log in	Indexes on users(username), submissions	schema.sql	Performance.LookupWithEightHundredStudents…
NFR-02	Log in, create user	SodiumPasswordHasher (Argon2id)	SodiumPasswordHasher.cpp	Authentication.Sodium…, PasswordsAreNeverStored…, ChangePassword…
NFR-03	All	Local SQLite, no network code	whole project	Inspection: no network dependencies
NFR-04	All writing use cases	IUnitOfWork, SqliteScope, foreign keys, CHECK constraints	Database.cpp, SqliteRepositories.cpp	Persistence., *Integrity.* (3), TypesCodes.*
NFR-05	All	Role menus in the console	ui/ConsoleApp.cpp	Inspection and walkthrough of the menus
NFR-06	All	Validation, exception hierarchy, bound parameters	Errors.hpp, Guards.hpp, Console.cpp	Injection., *TimeUtil.*, Integrity.DatabaseRejects…, Workflow.InvalidMarks…
NFR-07	n/a	CI pipeline, CMake warnings	CMakeLists.txt, ci.yml	Coverage report: 91.3% business logic

Table A.1: Traceability from requirements to use cases, design, source and tests.

Appendix B: Full Test Results

All 61 automated tests passed on the final build (GoogleTest, run through CTest). Business-logic coverage 91.3% (567 of 621 lines).

ID	Suite	Test case	Level	Requirements	Result
T-01	Penalty	Early Submission Is On Time	Unit	FR-10	Pass
T-02	Penalty	Submission At Deadline Instant Is On Time	Unit	FR-10	Pass
T-03	Penalty	One Second Late Counts As One Day	Unit	FR-10	Pass
T-04	Penalty	Exactly One Day Late Is Still One Day	Unit	FR-10	Pass
T-05	Penalty	One Day And One Second Is Two Days	Unit	FR-10	Pass
T-06	Penalty	Exactly Three Days Late Is Last Penalised Day	Unit	FR-10	Pass
T-07	Penalty	More Than Three Days Late Scores Zero	Unit	FR-10	Pass
T-08	Penalty	Apply Deducts Percentage Of Maximum Mark	Unit	FR-10	Pass
T-09	Penalty	Apply Never Goes Below Zero	Unit	FR-10	Pass
T-10	Penalty	On Time Work Keeps Raw Mark	Unit	FR-10	Pass
T-11	Penalty	Extended Deadline Removes Penalty	Unit	FR-07, FR-10	Pass
T-12	Penalty	Constructor Rejects Invalid Parameters	Unit	FR-10	Pass
T-13	Penalty	Strategy Is Replaceable Through Interface	Unit	FR-10	Pass
T-14	AccessControl	Roles Cannot Perform Other Roles Actions	Integration	FR-02	Pass
T-15	AccessControl	Lecturer Only Acts On Courses They Teach	Integration	FR-02, FR-07, FR-11	Pass
T-16	AccessControl	Head Of Department Only Moderates Own Department	Integration	FR-02, FR-12	Pass
T-17	AccessControl	Unenrolled Student Cannot Submit	Integration	FR-08	Pass
T-18	AccessControl	Release Requires Approval First	Integration	FR-13	Pass
T-19	AccessControl	Students See Only Their Own Released Marks	Integration	FR-13, FR-16	Pass
T-20	Authentication	Login Succeeds And Failures Share One Message	Integration	FR-01	Pass
T-21	Authentication	Deactivated Accounts Cannot Log In	Integration	FR-17	Pass
T-22	Authentication	Passwords Are Never Stored In Plain Text	Integration	NFR-02	Pass
T-23	Authentication	Sodium Hasher Produces Salted Argon2id Hashes	Integration	NFR-02	Pass
T-24	Authentication	Change Password Requires Current Password And AValid New One	Integration	FR-18, NFR-02	Pass
T-25	AuditTrail	Only Administrators And Heads Of Department May View History	Integration	FR-19	Pass
T-26	AdminRules	Account Validation	Integration	FR-03, FR-04	Pass
T-27	Injection	Sql Metacharacters Are Treated As Data	Integration	NFR-06	Pass
T-28	Integrity	Audit Log Cannot Be Edited Or Deleted	Integration	FR-14	Pass

ID	Suite	Test case	Level	Requirements	Result
T-29	Integrity	Database Rejects Invalid Rows	Integration	NFR-04, NFR-06	Pass
T-30	Integrity	Uncommitted Scope Rolls Back	Integration	NFR-04	Pass
T-31	Persistence	Data Survives Restart	Integration	NFR-04	Pass
T-32	Performance	Lookup With Eight Hundred Students Is Fast	Integration	NFR-01	Pass
T-33	TimeUtil	Parses And Formats Round Trip	Unit	NFR-06	Pass
T-34	StateMachine	Happy Path Reaches Released	Unit	FR-15	Pass
T-35	StateMachine	Changes Requested Loops Back To Review	Unit	FR-15	Pass
T-36	StateMachine	Resubmission Supersedes Earlier Attempt	Unit	FR-15	Pass
T-37	StateMachine	Marks Cannot Be Released Without Approval	Unit	FR-15	Pass
T-38	StateMachine	Terminal States Allow Nothing	Unit	FR-15	Pass
T-39	StateMachine	Illegal Transition Throws And Leaves Status Unchanged	Unit	FR-15	Pass
T-40	Types	String Codes Round Trip	Unit	FR-15, NFR-04	Pass
T-41	TypesCodes	All Enumerations Round Trip Through Their Database Codes	Unit	NFR-04	Pass
T-42	TypesCodes	Unknown Codes Are Rejected	Unit	NFR-06	Pass
T-43	User	Student May Submit And View Marks Only	Unit	FR-02	Pass
T-44	User	Lecturer Marks But Cannot Moderate Or Release	Unit	FR-02	Pass
T-45	User	Head Of Department Moderates And Releases	Unit	FR-02	Pass
T-46	User	Administrator Manages Accounts But Never Marks	Unit	FR-02	Pass
T-47	User	Roles Are Resolved Polymorphically	Unit	FR-02	Pass
T-48	User	Authentication Checks Password And Active Flag	Unit	FR-01, FR-17	Pass
T-49	Workflow	Mark Is Hidden From Student Until Hod Releases It	Integration	FR-12, FR-13, FR-15	Pass
T-50	Workflow	Changes Requested Loop Then Approval	Integration	FR-12, FR-14	Pass
T-51	Workflow	Changes Requested Needs AComment	Integration	FR-12	Pass
T-52	Workflow	Late Work Is Accepted Flagged And Penalised At Marking	Integration	FR-10, FR-11	Pass
T-53	Workflow	Work More Than Three Days Late Is Accepted But Scores Zero	Integration	FR-10	Pass
T-54	Workflow	Resubmission Before Deadline Supersedes Earlier Attempt	Integration	FR-09	Pass
T-55	Workflow	Resubmission After Deadline Is Refused	Integration	FR-09	Pass
T-56	Workflow	Attempt Limit Is Enforced	Integration	FR-06, FR-09	Pass
T-57	Workflow	Marked Work Cannot Be Replaced By Resubmission	Integration	FR-09	Pass
T-58	Workflow	Extension Shifts The Students Deadline	Integration	FR-07	Pass
T-59	Workflow	Extension Must Move The Deadline Later And Needs AReason	Integration	FR-07	Pass

ID	Suite	Test case	Level	Requirements	Result
T-60	Workflow	Mark Changes Are Audited With Old New Reason And Actor	Integration	FR-11, FR-14	Pass
T-61	Workflow	Invalid Marks Are Rejected	Integration	FR-11	Pass

Table B.1: Automated test cases, level, requirements covered and results.

Appendix C: User Manuals

Common to all users. Start the program and enter your username and password at the login screen. Enter 0 to exit. After three wrong attempts you can keep trying, but the system never says whether the username or the password was wrong. Choose menu options by typing the number and pressing Enter. Type 0 at a menu to log out. Use “Change my password” regularly; passwords must have at least 8 characters. If you see an error message, read it and try again; nothing has been changed when an error appears.

Student. *My courses and assignments* lists your courses, deadlines, maximum marks and how many attempts you have used. *Submit work*: choose the assignment, then type the path of your file. The system tells you if the work is late and what penalty applies. You may resubmit before the deadline, until you reach the attempt limit, but not once your work has been marked. *My submissions* shows each attempt and its status (Submitted, Being assessed, Result available). *My marks and feedback* shows only results that the Head of Department has released, with written feedback; if you have none yet the system says so.

Lecturer. *My courses and assignments* shows the courses you teach. *Create assignment*: choose the course, then enter a title, description, deadline (format YYYY-MM-DD HH:MM), maximum mark and attempt limit. *Mark submissions*: choose the assignment and a submission, then enter a mark and written feedback. If you are changing an existing mark you must also give a reason, which is recorded. The mark then goes to your Head of Department for moderation. *Grant an extension*: choose the assignment and student, enter the new deadline and a reason. In v1.1 you can also grant a whole class the same extension.

Head of Department. You have the lecturer menu and three more options. *Moderate marks* lists marks in your department waiting for review. Approve a mark, or send it back to the lecturer with a comment. *Release approved marks* makes approved marks visible to students; release one, or all. *View audit history* shows who changed a mark or record, when, and why.

Administrator. On the very first start the system asks you to create the initial administrator. *Create user account* creates students, lecturers, Heads of Department and administrators (a department must exist first for staff). *List users* and *Activate or deactivate a user* manage accounts; deactivated users cannot log in. *Create department*, *Create course*, *Assign lecturer to course* and *Enrol student in course* set up teaching. *View audit history* shows the history of any record. Back up the database file daily (see the installation guide).

Appendix D: Updated Risk Register

ID	Risk	Initial exposure	Status at final gate	Residual exposure	Notes
R1	Schedule overrun	16	Reduced	6	Must requirements built first; console interface kept
R2	Requirements conflicts cause rework	12	Closed	2	All conflicts resolved in the SRS before design
R3	Security flaw exposes passwords or marks	10	Reduced	4	Argon2id, bound parameters, role and ownership checks, injection tests
R4	Poor performance on old computers	9	Reduced	3	800-user lookup measured; hashing cost tuned
R5	Data loss or corruption	10	Reduced	4	Transactions, constraints, backup procedure; depends on school discipline
R6	Build or library problems on target	9	Reduced	4	CI build, installation guide
R7	Developer unavailability	8	Accepted	4	Buffer used in week seven
R8	Scope creep	9	Closed	2	Out-of-scope list enforced; CR1 handled through change process
R9	New: Argon2id memory use on very old computers	6	Open	4	Interactive parameters can be lowered in one constant if needed
R10	New: lateness caused by school internet outages treated as student fault	6	Open	4	Individual and class-wide extensions; school should review policy

Table D.1: Risk register at the final gate.

Appendix E: SQL Schema

```sql
-- LMS database schema (SQLite). Normalised to 3NF; see report section 5.4.
-- Timestamps are INTEGER seconds since the Unix epoch (UTC).
PRAGMA foreign_keys = ON;

CREATE TABLE IF NOT EXISTS departments (
    department_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name          TEXT    NOT NULL UNIQUE,
    hod_user_id   INTEGER UNIQUE REFERENCES users(user_id) ON DELETE SET NULL
);

CREATE TABLE IF NOT EXISTS users (
    user_id        INTEGER PRIMARY KEY AUTOINCREMENT,
    username       TEXT    NOT NULL UNIQUE COLLATE NOCASE,
    name           TEXT    NOT NULL,
    password_hash  TEXT    NOT NULL,          -- Argon2id string; the salt is embedded
    role           TEXT    NOT NULL CHECK (role IN ('ADMIN','LECTURER','HOD','STUDENT')),
    student_number TEXT    UNIQUE,
    staff_number   TEXT    UNIQUE,
    department_id  INTEGER REFERENCES departments(department_id),
    active         INTEGER NOT NULL DEFAULT 1 CHECK (active IN (0,1)),
    CHECK ((role = 'STUDENT' AND student_number IS NOT NULL AND staff_number IS NULL
                AND department_id IS NULL)
        OR (role IN ('LECTURER','HOD') AND staff_number IS NOT NULL AND student_number IS
NULL
                AND department_id IS NOT NULL)
        OR (role = 'ADMIN' AND student_number IS NULL AND staff_number IS NULL
                AND department_id IS NULL))
);

CREATE TABLE IF NOT EXISTS courses (
    course_id     INTEGER PRIMARY KEY AUTOINCREMENT,
    course_code   TEXT    NOT NULL UNIQUE,
    title         TEXT    NOT NULL,
    academic_year INTEGER NOT NULL CHECK (academic_year BETWEEN 2000 AND 2100),
    department_id INTEGER NOT NULL REFERENCES departments(department_id)
);

CREATE TABLE IF NOT EXISTS course_lecturers (
    course_id   INTEGER NOT NULL REFERENCES courses(course_id) ON DELETE CASCADE,
    lecturer_id INTEGER NOT NULL REFERENCES users(user_id),
    PRIMARY KEY (course_id, lecturer_id)
);

CREATE TABLE IF NOT EXISTS enrolments (
    enrolment_id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id   INTEGER NOT NULL REFERENCES users(user_id),
    course_id    INTEGER NOT NULL REFERENCES courses(course_id) ON DELETE CASCADE,
    enrolled_at  INTEGER NOT NULL,
    status       TEXT    NOT NULL DEFAULT 'ACTIVE' CHECK (status IN
('ACTIVE','WITHDRAWN')),
    UNIQUE (student_id, course_id)
);

CREATE TABLE IF NOT EXISTS assignments (
    assignment_id INTEGER PRIMARY KEY AUTOINCREMENT,
    course_id     INTEGER NOT NULL REFERENCES courses(course_id) ON DELETE CASCADE,
    created_by    INTEGER NOT NULL REFERENCES users(user_id),
    title         TEXT    NOT NULL,
    description   TEXT    NOT NULL DEFAULT '',
    due_at        INTEGER NOT NULL,
    max_mark      REAL    NOT NULL CHECK (max_mark > 0),
    max_attempts  INTEGER NOT NULL DEFAULT 3 CHECK (max_attempts >= 1)
```

```
);

CREATE TABLE IF NOT EXISTS submissions (
    submission_id INTEGER PRIMARY KEY AUTOINCREMENT,
    assignment_id INTEGER NOT NULL REFERENCES assignments(assignment_id) ON DELETE CASCADE,
    student_id    INTEGER NOT NULL REFERENCES users(user_id),
    attempt_no    INTEGER NOT NULL CHECK (attempt_no >= 1),
    submitted_at  INTEGER NOT NULL,
    file_path     TEXT    NOT NULL,
    status        TEXT    NOT NULL CHECK (status IN ('DRAFT','SUBMITTED','UNDER_REVIEW',
                      'MARKED_PENDING_MODERATION','CHANGES_REQUESTED','APPROVED',
                      'RELEASED','SUPERSEDED')),
    is_late       INTEGER NOT NULL DEFAULT 0 CHECK (is_late IN (0,1)),
    UNIQUE (assignment_id, student_id, attempt_no)
);

CREATE TABLE IF NOT EXISTS extensions (
    extension_id     INTEGER PRIMARY KEY AUTOINCREMENT,
    assignment_id    INTEGER NOT NULL REFERENCES assignments(assignment_id) ON DELETE
CASCADE,
    student_id       INTEGER NOT NULL REFERENCES users(user_id),
    requested_due_at INTEGER NOT NULL,
    reason           TEXT    NOT NULL,
    status           TEXT    NOT NULL DEFAULT 'PENDING'
                         CHECK (status IN ('PENDING','APPROVED','REJECTED')),
    approved_by      INTEGER REFERENCES users(user_id)
);

CREATE TABLE IF NOT EXISTS marks (
    mark_id       INTEGER PRIMARY KEY AUTOINCREMENT,
    submission_id INTEGER NOT NULL UNIQUE REFERENCES submissions(submission_id) ON DELETE
CASCADE,
    marked_by     INTEGER NOT NULL REFERENCES users(user_id),
    raw_mark      REAL    NOT NULL CHECK (raw_mark >= 0),
    final_mark    REAL    NOT NULL CHECK (final_mark >= 0),
    feedback      TEXT    NOT NULL DEFAULT '',
    status        TEXT    NOT NULL CHECK (status IN
('PENDING_MODERATION','CHANGES_REQUESTED',
                      'APPROVED','RELEASED')),
    marked_at     INTEGER NOT NULL,
    released_at   INTEGER,
    CHECK (final_mark <= raw_mark)
);

CREATE TABLE IF NOT EXISTS moderation_records (
    moderation_id INTEGER PRIMARY KEY AUTOINCREMENT,
    mark_id       INTEGER NOT NULL REFERENCES marks(mark_id) ON DELETE CASCADE,
    reviewed_by   INTEGER NOT NULL REFERENCES users(user_id),
    decision      TEXT    NOT NULL CHECK (decision IN ('APPROVED','CHANGES_REQUESTED')),
    comment       TEXT    NOT NULL DEFAULT '',
    decided_at    INTEGER NOT NULL
);

CREATE TABLE IF NOT EXISTS audit_log (
    audit_id      INTEGER PRIMARY KEY AUTOINCREMENT,
    actor_user_id INTEGER REFERENCES users(user_id) ON DELETE SET NULL,
    action        TEXT    NOT NULL,
    entity_type   TEXT    NOT NULL,
    entity_id     INTEGER NOT NULL,
    old_value     TEXT,
    new_value     TEXT,
    reason        TEXT,
    occurred_at   INTEGER NOT NULL
);
```

```sql
-- The audit trail is append-only: no edits, no deletions.
CREATE TRIGGER IF NOT EXISTS audit_log_no_update BEFORE UPDATE ON audit_log
BEGIN
    SELECT RAISE(ABORT, 'audit_log is append-only');
END;

CREATE TRIGGER IF NOT EXISTS audit_log_no_delete BEFORE DELETE ON audit_log
BEGIN
    SELECT RAISE(ABORT, 'audit_log is append-only');
END;

CREATE INDEX IF NOT EXISTS idx_enrolments_course     ON enrolments(course_id);
CREATE INDEX IF NOT EXISTS idx_assignments_course    ON assignments(course_id);
CREATE INDEX IF NOT EXISTS idx_submissions_lookup    ON submissions(assignment_id,
student_id);
CREATE INDEX IF NOT EXISTS idx_extensions_lookup     ON extensions(assignment_id,
student_id);
CREATE INDEX IF NOT EXISTS idx_moderation_mark       ON moderation_records(mark_id);
CREATE INDEX IF NOT EXISTS idx_audit_entity          ON audit_log(entity_type, entity_id);
```

Appendix F: Additional Diagrams and Specifications

F.1 Use case model.

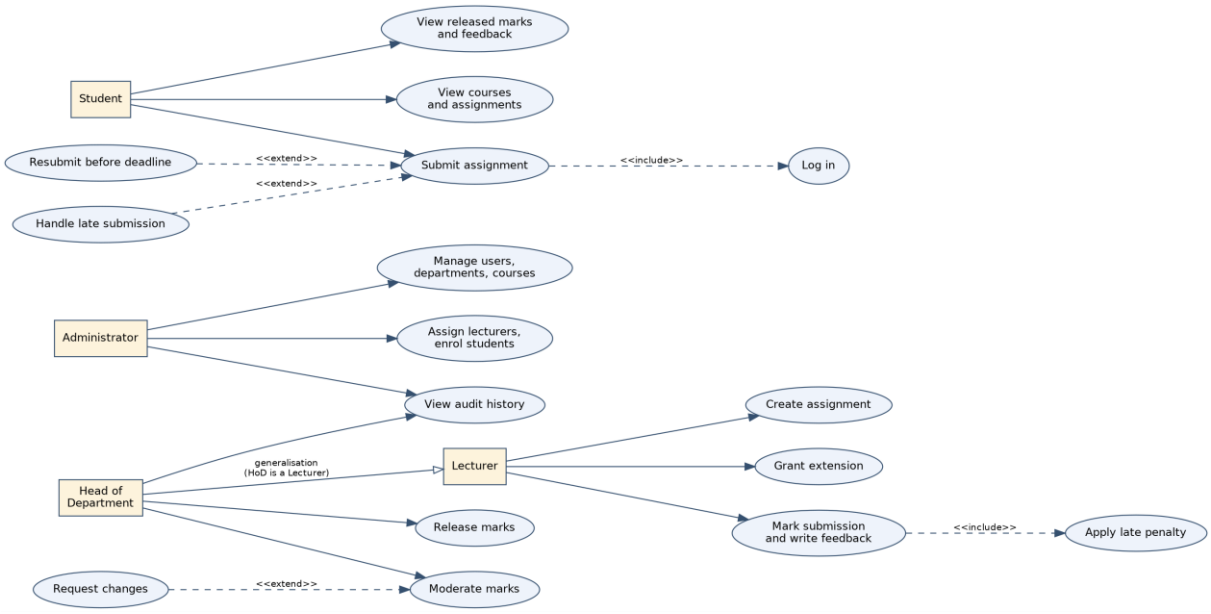


Figure F1: Use case diagram for the four actors, with include, extend and generalisation relationships.

UC-02 Mark submission	
Actor	Lecturer (or HoD acting as lecturer)
Preconditions	The lecturer is logged in and teaches the course. The submission is Submitted, Under review or Changes requested.
Main flow	1. The lecturer selects an assignment and a submission. 2. The lecturer enters a mark and written feedback. 3. The system checks that the mark is between 0 and the maximum. 4. The system computes the late penalty from the submission time and effective deadline. 5. In one transaction the system moves the submission to Marked pending moderation, creates the mark with status Pending moderation and writes an audit entry.
Alternative flows	A1 Revising a mark returned by the HoD: the lecturer must give a reason; the system updates the existing mark and records old value, new value and reason.
Exception flows	E1 Lecturer does not teach the course: access denied. E2 Mark out of range or feedback empty: validation error. E3 Submission already marked or released: refused.
Postconditions	The mark awaits moderation and is not visible to the student.
Business rules	BR1 Feedback is mandatory. BR2 Every mark change is audited with a reason. BR3 The final mark is the raw mark after the late penalty.

Table F1: Use case specification UC-02.

UC-03 Moderate and release marks	
Actor	Head of Department
Preconditions	The HoD is logged in and heads the department that owns the course. A mark is waiting for moderation.
Main flow	1. The HoD opens the marks waiting for moderation in the department. 2. The HoD reviews one mark and its feedback. 3. The HoD approves it; the system sets the mark and submission to Approved and records a moderation record and audit entry. 4. Later the HoD releases approved marks; the system sets Released, stores the release time and writes an audit entry. 5. The student can now see the mark and feedback.

UC-03 Moderate and release marks	
Alternative flows	A1 Changes requested: the HoD returns the mark with a mandatory comment; the submission goes to Changes requested and the lecturer revises it.
Exception flows	E1 The mark belongs to another department: access denied. E2 The mark is not awaiting moderation, or not approved when releasing: refused. E3 Missing comment when returning: validation error.
Postconditions	Approved and released marks are visible only after the release step. Every decision is in the moderation records and audit log.
Business rules	BR1 Approval and release are separate steps. BR2 Only released marks are visible to students.

Table F2: Use case specification UC-03.

F.2 Architecture and behavioural models.

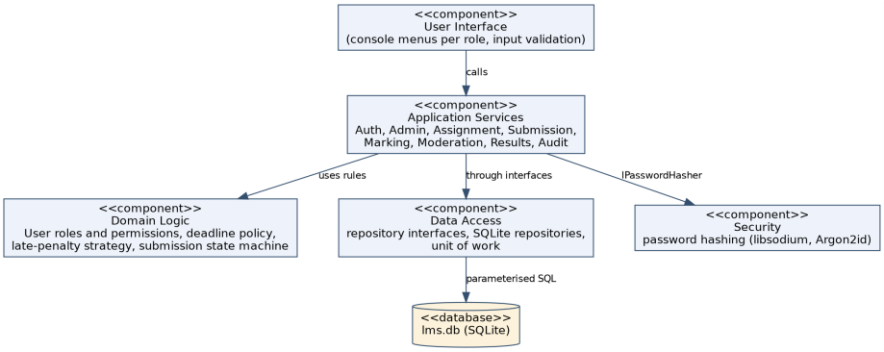


Figure F2: Component diagram of the layered architecture.

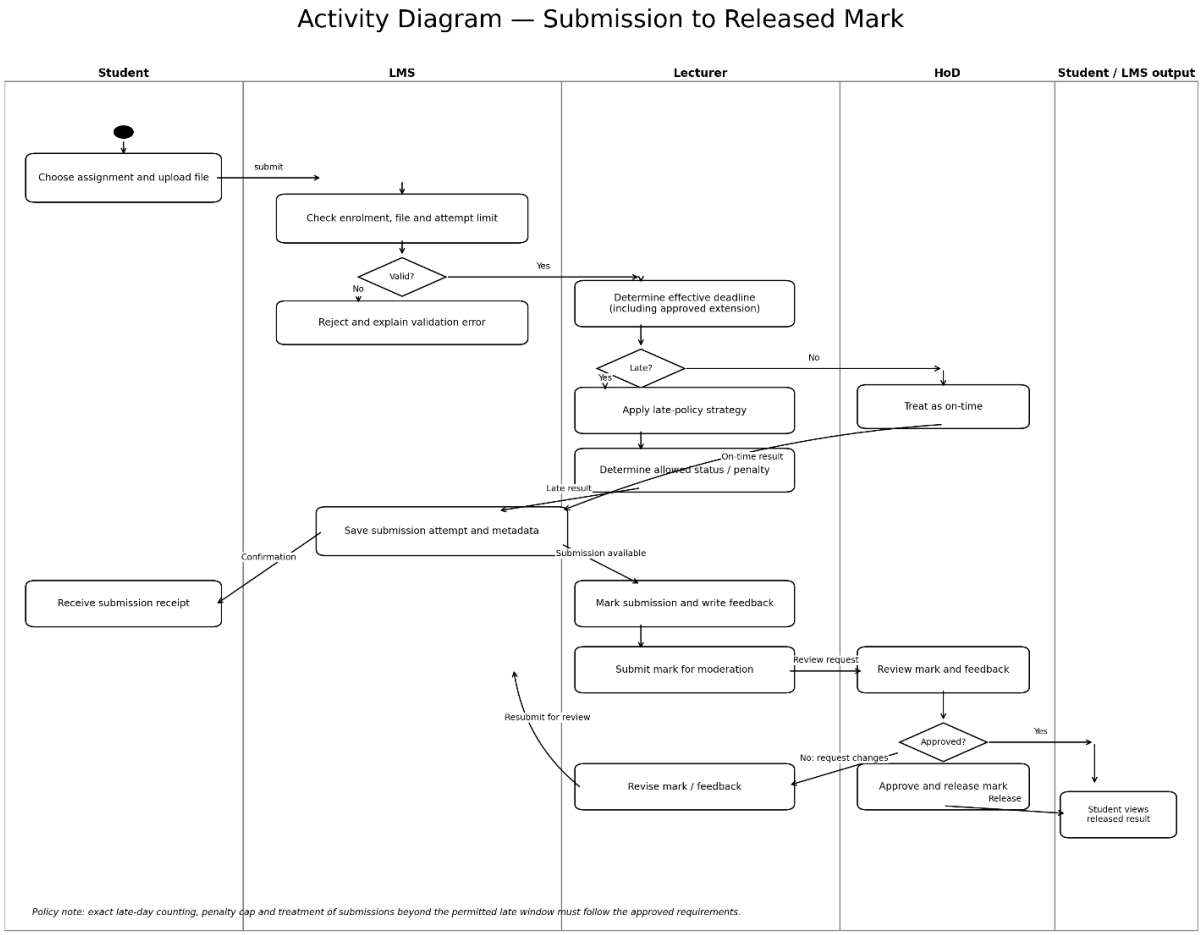


Figure F3: Activity diagram with swimlanes, from submission to released mark.

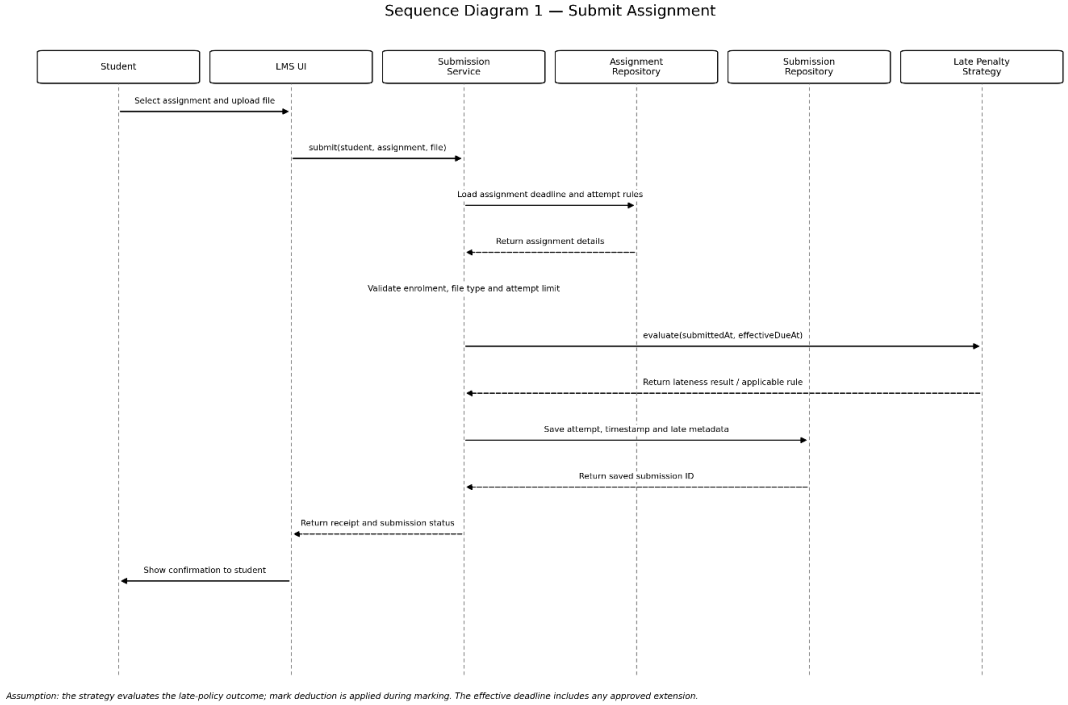


Figure F4: Sequence diagram for Submit assignment, including late and resubmission paths.

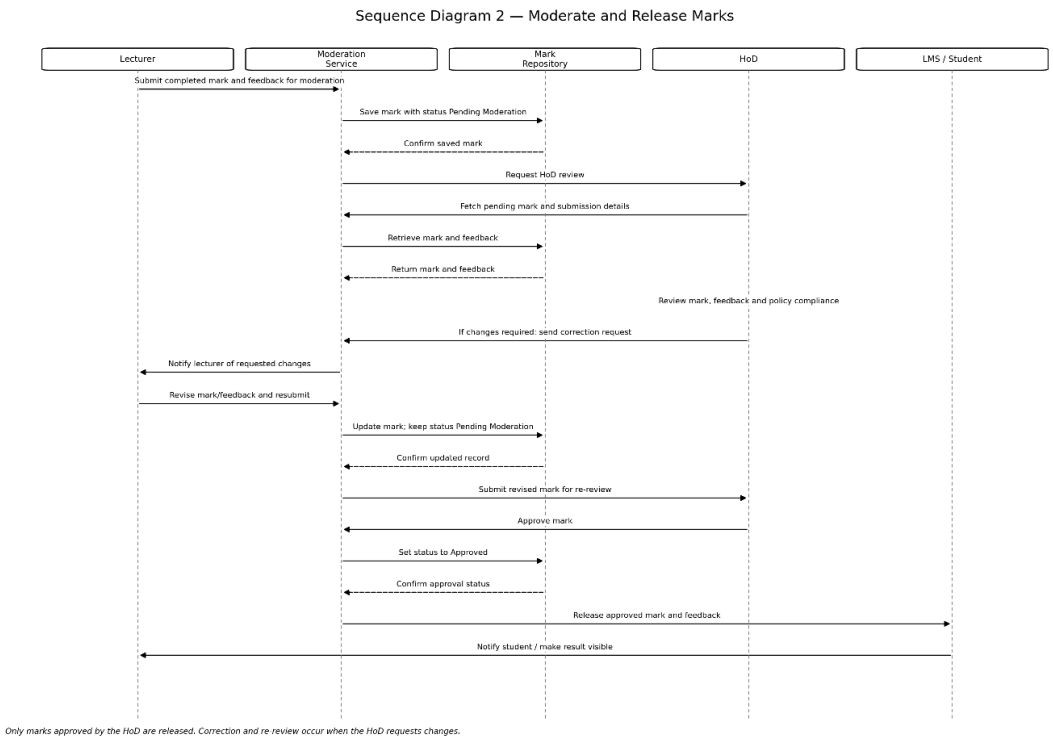


Figure F5: Sequence diagram for Moderate and release marks.

F.3 Database design.

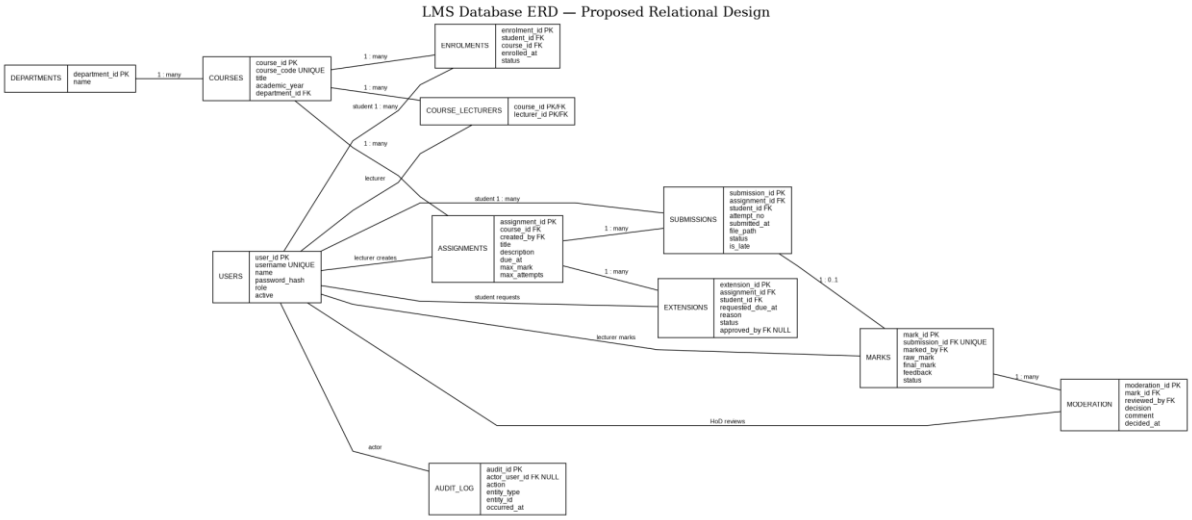


Figure F6: Entity-relationship diagram.

Table	Key columns and attributes	Constraints and notes
departments	department_id PK, name, hod_user_id FK	name unique; one HoD per department
users	user_id PK, username, name, password_hash, role, student_number, staff_number, department_id FK, active	username unique, case-insensitive; role in ADMIN, LECTURER, HOD, STUDENT; role-specific CHECK on numbers and department
courses	course_id PK, course_code, title, academic_year, department_id FK	code unique
course_lecturers	course_id FK, lecturer_id FK (composite PK)	many-to-many junction

Table	Key columns and attributes	Constraints and notes
enrolments	enrolment_id PK, student_id FK, course_id FK, enrolled_at, status	unique (student, course)
assignments	assignment_id PK, course_id FK, created_by FK, title, description, due_at, max_mark, max_attempts	max_mark > 0; attempts 1 or more
submissions	submission_id PK, assignment_id FK, student_id FK, attempt_no, submitted_at, file_path, status, is_late	unique (assignment, student, attempt); status limited to the eight states
extensions	extension_id PK, assignment_id FK, student_id FK, requested_due_at, reason, status, approved_by FK	status PENDING, APPROVED, REJECTED
marks	mark_id PK, submission_id FK unique, marked_by FK, raw_mark, final_mark, feedback, status, marked_at, released_at	final_mark at most raw_mark; one mark per submission
moderation_records	moderation_id PK, mark_id FK, reviewed_by FK, decision, comment, decided_at	history of every HoD decision
audit_log	audit_id PK, actor_user_id FK, action, entity_type, entity_id, old_value, new_value, reason, occurred_at	append-only: triggers abort UPDATE and DELETE

Table F3: Data dictionary summary.

F.4 Deployment and release notes.

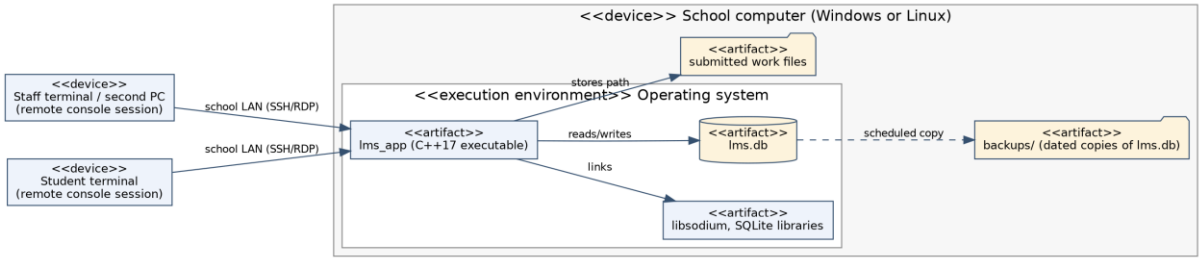


Figure F7: UML deployment diagram.

Version	Contents
v1.0	All Must and Should requirements FR-01 to FR-19 and NFR-01 to NFR-07: four roles, assignments, individual extensions, submission and resubmission, late penalty, marking with feedback, HoD moderation and release, audit log, Argon2id passwords, SQLite persistence, console interface, CI pipeline, 61 automated tests.
Known limits	One computer or remote sessions only; submitted files are referenced by path, not copied; individual extensions only.

Table F4: Release notes for v1.0.